

Feedback Loops

Project: Transparent Book Recommender

1. Feedback Loops

Teach the learning cycle

Primary target: Implement a deterministic recommender

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/24, ai-2/33. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a weighted score

Reference example, not the only acceptable wording: Score 8.

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A recommender shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

Concept to explain

Recommendations influence behavior, which then becomes new data.

Term · Meaning

Term: feedback loop

Meaning: A cycle in which outputs affect later inputs and therefore future outputs.

Term: exposure

Meaning: The opportunity to see or encounter an item, which can affect later choices.

Term: interaction data

Meaning: Records of what people were shown and how they acted, such as views, clicks or corrections.

Term: narrowing

Meaning: Reducing the allowed choices, scope or interpretation of a task.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 24 as a two-session investigation using fictional data. Build a fictional activity recommender using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name. This makes the outcome reproducible rather than dependent on input order. Assume unique names and validated numeric fields. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No. Filter by the hard constraint before ranking.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can explain how recommendation exposure can create a feedback loop.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. System recommends mostly fantasy.
2. Reader can only click among shown books.
3. Clicks are treated as stronger fantasy preference.
4. Next round shows even more fantasy, hiding other possible interests.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A recommender shows only adventure books, uses clicks to learn and then shows more adventure.

Explain the feedback loop.

Exact explanation to model

Exposure affects clicks; clicks affect future exposure. Record what was shown and offer controlled variety.

Explain the added worked example

Build a fictional activity recommender using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name. This makes the outcome reproducible rather than dependent on input order. Assume unique names and validated numeric fields.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run. After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

Key terms

Term · Meaning

Term: Recommender

Meaning: A procedure that suggests options using a stated policy

Term: Tie

Meaning: Two options have the same comparison score

Term: Deterministic

Meaning: The same inputs and rules produce the same result

Term: Explanation

Meaning: Evidence describing how a result was obtained

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Predict the result if the list order is reversed, and if the limit is 8.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Feedback Loops

A fictional catalogue repeatedly recommends two books using observed clicks.

The facts we will use

Initially each click count is 0. Equal counts choose earlier ID A over B.

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.

AI ACADEMY / AI-2/24

Exposure and clicks after the invented three rounds

	Shown	Clicked
A	3	3
B	0	0

Read a cell using both its row and column labels.

Feedback Loops

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial tie

Counts $A=0, B=0$; ID rule shows A.



2. Record event

A receives a click, so its count becomes 1; B stays 0.



3. Repeat selection

Higher A count causes A to be shown again; after three clicks $A=3, B=0$.



4. Interpret carefully

B's zero may mean no exposure, not dislike.

CHANGE ONE THING

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Initial tie

What happens and why: Counts $A=0, B=0$; ID rule shows A.

Step: 2. Record event

What happens and why: A receives a click, so its count becomes 1; B stays 0.

Step: 3. Repeat selection

What happens and why: Higher A count causes A to be shown again; after three clicks $A=3, B=0$.

Step: 4. Interpret carefully

What happens and why: B's zero may mean no exposure, not dislike.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The selection policy changes which evidence is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books. Keep display counts separate from click counts; investigate alternatives with a teacher-approved exposure plan rather than treating zero as negative feedback.

What this example does not establish

A click can reflect curiosity or a mistake; even equal exposure does not reveal every reason for selection.

Change one thing in the pictured case

Change: On a planned trial, show B once and record no click.

Prediction: B gains one exposure but still zero clicks.

Reason: The observation now supports a nonclick on that display, not a permanent dislike.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A recommender shows only adventure books, uses clicks to learn and then shows more adventure.

Write the question to explain: Explain the feedback loop.

Reveal after predictions: Exposure affects clicks; clicks affect future exposure. Record what was shown

and offer controlled variety.

Keep for an independent attempt: Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Reversing still returns A. Limit 8 leaves no eligible item and returns None. Exit: X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: State `recommend([], 10)` and the worked items with limit 8.

Teacher check: Both return `None`; the caller should display no eligible option.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What extra field would reveal why comparing raw clicks is misleading?

Reasoned teacher answer

Display or exposure count: A shown three times, B zero. Raw clicks confound opportunities with response.

Changed-case reasoning: B gains one exposure but still zero clicks. The observation now supports a nonclick on that display, not a permanent dislike.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Act out three recommendation rounds. Log shown items, choices, and updated profile; add an exploration rule and compare diversity.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

A convenient shortcut preserves meaning: Updating best on score $\geq$ best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Exposure affects clicks; clicks affect future exposure. Record what was shown and offer controlled variety.

Independent question

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

Expected independent answer

No. Exposure affects what can be clicked. Showing space produces space clicks that may cause even more space recommendations; test other relevant options to learn more.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Clicks reveal preferences independently of what is offered.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Poetry is $1/2=50\%$; adventure is $6/20=30\%$. Raw clicks favour adventure but exposure rates differ. Position, user mix and tiny poetry sample confound interpretation. Run a balanced comparison and record exposures and outcomes.

Explanations for the additional workbook examples

W1 Trace selection

Why does A win the worked example even though B is processed first?

Both score 8 and qualify; A is alphabetically first under the tie rule.

W2 Handle no match

State `recommend([], 10)` and the worked items with limit 8.

Both return `None`; the caller should display no eligible option.

For the empty list, the loop has no iterations, so best stays `None`. With the worked durations 10 and 9 under a limit of 8, neither item passes the eligibility check, so best again stays `None`. The caller must turn this result into a clear no-eligible-option message; it must not treat `None` as an activity name.

W3 Repair order dependence

A tied selection changes after reversing input order. Identify the likely missing policy and test.

An explicit deterministic tie rule is missing or incorrect. Apply alphabetical name order and rerun both orders.

W4 Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns `None`.

W5 Explain the result

Write a one-sentence explanation of X winning without calling it objectively best.

X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Assess independent transfer

Selected case: Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

Selected case answer

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns `None`.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Reserve some varied suggestions and distinguish 'not shown' from 'not liked.'

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 25: Text as Pieces. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Trace best and best_score row by row. Extension: return a record with name, score and reason while retaining the same selection policy.

Independent W4 evidence: X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None. Record whether the learner needed no help, a hint or explanation. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

Next week: File input and output. Reuse this week's explicit contracts and evidence.

Plan the delayed return

Suggested return points in this level: ai-2/25, ai-2/27, ai-2/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive

project does not by itself demonstrate understanding.

Project evidence from this case

Log recommender exposure, choice and explicit feedback separately; consider whether its own policy narrows future evidence.

7. Prepare, teach and assess

Teaching target: Implement a deterministic recommender

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: `State recommend([], 10)` and the worked items with limit 8.

Reasoned practice answer: Both return None; the caller should display no eligible option.

Check every learner before discussion: Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: On a planned trial, show B once and record no click. Require a prediction and a reason before comparing. Expected result: B gains one exposure but still zero clicks. Reason: The observation now supports a nonclick on that display, not a permanent dislike.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.